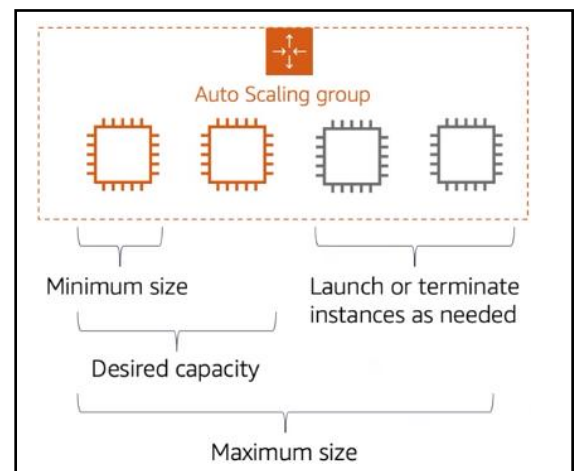
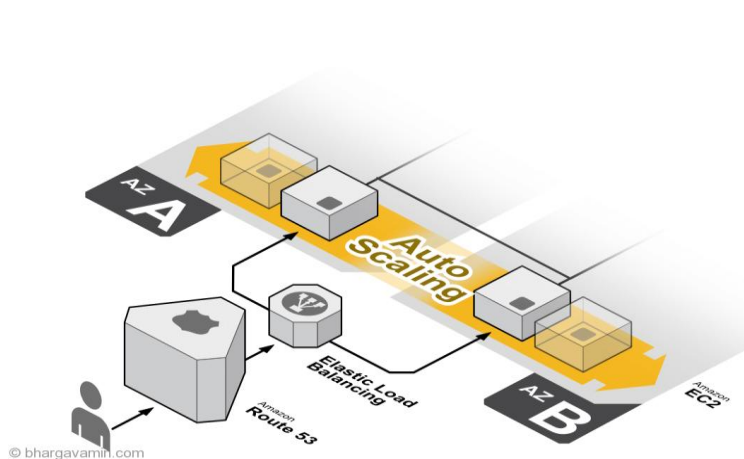


Auto Scaling

Class no:23

Auto Scaling in AWS

- **Definition:** Auto Scaling dynamically adds or removes compute resources (like EC2 instances) based on traffic or workload.
- **Purpose:** Ensures applications remain available, responsive, and cost-efficient.
- **Core Component:** Auto Scaling Groups (ASGs) — collections of EC2 instances managed together.
- An **Auto Scaling Group** is a logical grouping of EC2 instances managed together.
- ASGs ensure that applications remain available by automatically launching or terminating instances based on scaling policies. It defines:
 - **Minimum size** → the least number of instances always running.
 - **Maximum size** → the upper limit of instances allowed.
 - **Desired capacity** → the target number of instances at a given time.



Key Benefits

- **High Availability:** Maintains application uptime by replacing unhealthy instances.
- **Cost Optimization:** Only pays for resources when needed.
- **Flexibility:** Supports multiple scaling strategies (manual, scheduled, dynamic, predictive).
- **Integration:** Works with Elastic Load Balancing and CloudWatch for monitoring and distribution.

Types of Auto Scaling in AWS

Type	Description	Use Case
Dynamic Scaling	Adjusts capacity based on real-time metrics (CPU, memory, request count). Uses CloudWatch alarms and scaling policies.	E-commerce site handling traffic spikes during sales.
Predictive Scaling	Uses machine learning to forecast demand and scale resources ahead of time.	Applications with predictable traffic patterns (e.g., daily peak at 9 AM).
Scheduled Scaling	Scales resources at fixed times (e.g., increase instances at 8 AM, reduce at 10 PM).	Workloads with known schedules, like office-hour apps.
Manual Scaling	User manually sets desired capacity, minimum, and maximum instances.	Testing environments or controlled workloads.

Hands on creating an autoscaling group:

Step no:1 Go to instances → scroll down → Launch template,

Name:My template

OS: amzon linux

Instance type: T3 micro

Keypair: our keypair

Network settings,

Select existing group

Default

Create launch template.

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required

My-template

Must be unique to this account. Max 128 chars. No spaces or special characters like "%", "*", "!", "@".

Template version description

My-template

Max 255 chars

Auto Scaling guidance

[Info](#)

Select this if you intend to use this template with EC2 Auto Scaling.

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags

► Source template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start



[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-048f445314bcaa09 (64-bit (x86), uefi-preferred) / ami-07efda20bfc979493 (64-bit (ARM), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description
Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260406.2 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-048f445314bcaa09

Publish Date

2026-04-06

Username

ec2-user

Verified provider

▼ Instance type [Info](#) [Get advice](#)

Advanced

Instance type

t3.micro

Family: t3

2 vCPU

1 GiB Memory

Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour

On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

On-Demand RHEL base pricing: 0.04 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour

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On-Demand Windows base pricing: 0.0204 USD per Hour

Free tier eligible

All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

mumpk

[Create new key pair](#)

▼ Network settings [Info](#)

Subnet

Don't include in launch template

When you specify a subnet, a network interface is automatically added to your template.

[Create new subnet](#)

Availability Zone

Don't include in launch template

[Enable additional zones](#)

Firewall (security groups)

[Info](#)

A security group is a set of firewall rules that control the traffic for your instances. Add rules to allow specific traffic to reach your instances.

☒ Select existing security group

☐ Create security group

Security groups

[Info](#)

Select security groups

default sg-06d80e515c32b9515

VPC: vpc-0538c231a226fe3c9

[Compare security group rules](#)

► Advanced network configuration

Step no:2 Go to instances page → scroll down → Choose Auto scaling group,

Name: my-autoscaling-group

Launch template: Choose our created template

Choose launch template [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name

Enter a name to identify the group.

my-auto-scaling-group

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

My-template

[Create a launch template](#)

Version

Default (1)

[Create a launch template version](#)

Description

My-template

Launch template

[My-template](#)

lt-0bb50995d1228f1d5

Instance type

t3.micro

AMI ID

ami-048f4445314bcaa09

Security groups

-

Request Spot Instances

No

Key pair name

mumkp

Security group IDs

[sg-06d80e515c32b9515](#)

Additional details

Storage (volumes)

-

Date created

Fri Apr 10 2026 18:39:59 GMT+0530 (India Standard Time)

[Cancel](#)

[Next](#)

Network settings,

Choose Default VPC

Choose All subnets (for high availability)

Choose balanced best effect.

Choose instance launch options [Info](#)

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements [Info](#)

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template

[My-template](#)

lt-0bb50995d1228f1d5

Version

Default

Description

My-template

Instance type

t3.micro

[Override launch template](#)

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-0538c231a226fe3c9

172.31.0.0/16 Default

[Create a VPC](#)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

aps1-az1 (ap-south-1a) | subnet-01bf4d440c0912db5

172.31.32.0/20 Default

aps1-az2 (ap-south-1d) | subnet-0c305b80f72541592

172.31.16.0/20 Default

aps1-az3 (ap-south-1b) | subnet-0b3c21b6de4314d95

172.31.0.0/20 Default

[Create a subnet](#)

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

Balanced best effort

If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

Balanced only

If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

[Cancel](#)

[Skip to review](#)

[Previous](#)

[Next](#)

Step no:3 We will attach the load balancer & its target groups there. We attach the Load Balancer with respective TG & Ec2.

Choose: No Load balancer

Health check: 300

Keep the rest as default → click next.

Integrate with other services - optional [Info](#)

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

☒ **No load balancer**
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ **Attach to an existing load balancer**
Choose from your existing load balancers.

☐ **Attach to a new load balancer**
Quickly create a basic load balancer to attach to your Auto Scaling group.

VPC Lattice integration options [Info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services to AWS.

Select VPC Lattice service to attach

☒ **No VPC Lattice service**
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ **Attach to VPC Lattice service**
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#)

Application Recovery Controller (ARC) zonal shift - new [Info](#)

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ **Enable zonal shift**
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

☒ **Always enabled**

Additional health check types - optional [Info](#)

☐ **Turn on Elastic Load Balancing health checks**
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ **Turn on VPC Lattice health checks**
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ **Turn on Amazon EBS health checks**
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [Info](#)

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

seconds

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

Step no:4 Configure the group size → Desired capacity:2 & Min. desired capacity:1 & Max. desired capacity:3

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Desired capacity

Specify your group size.

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

Equal or less than desired capacity

Max desired capacity

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ **No scaling policies**
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ **Target tracking scaling policy**
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Meet behavior

☒ **No policy**
For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Prioritize availability

☐ **Launch before terminating**
Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☐ **Terminate and launch**
Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

☐ **Custom behavior**
Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

Additional capacity settings

Capacity Reservation preference [Info](#)

Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.

☒ **Default**
Auto Scaling uses the Capacity Reservation preference from your launch template.

Step no:5 Click next → next → Create auto scaling group.

Step 5: Add notifications Edit

Notifications

No notifications

Step 6: Add tags Edit

Tags (0)

Key	Value	Tag new instances
No tags		

[Preview code](#) [Cancel](#) [Previous](#) [Create Auto Scaling group](#)

Step no:6 Once created, move to **Automatic Scaling Tab** → click **create dynamic scaling policy**

my-auto-scaling-group

my-auto-scaling-group Capacity overview Edit

arn:aws:autoscaling:ap-south-1:314383175192:autoScalingGroup:cb1ec617-8233-4430-a3b7-0ef3a60a7f44:autoScalingGroupName/my-auto-scaling-group

Desired capacity 2	Scaling limits 1 - 3	Desired capacity type Units (number of instances)	Status Updating capacity
------------------------------	--------------------------------	---	---

Date created
Fri Apr 10 2026 18:43:51 GMT+0530 (India Standard Time)

[Details](#) [Integrations](#) [Automatic scaling](#) [Instance management](#) [Instance refresh](#) [Activity](#) [Monitoring](#) [Tags](#)

Scaling policies resize your Auto Scaling group to meet changes in demand. With reactive dynamic scaling policies, you can track specific CloudWatch metrics and take action when the CloudWatch alarm threshold is met. Use predictive scaling policies along with dynamic scaling policies in the following situations: when your application demand changes quickly, but with a recurring pattern, or when your EC2 instances require more time to initialize.

Dynamic scaling policies (0) [Info](#) [Actions](#) [Create dynamic scaling policy](#)

No dynamic scaling policies have been created

Dynamic scaling policies use real-time data to scale your group based on configurable metrics.

[Create dynamic scaling policy](#)

Choose the metric type as: **Average CPU utilization**, and target value as **1** (Normally target value ranges between 60-80 %)

Create dynamic scaling policy

Policy type
Target tracking scaling

Scaling policy name
Target Tracking Policy

Metric type [Info](#)
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.
Average CPU utilization

Target value
1

Instance warmup [Info](#)
10 seconds

☐ Disable scale in to create only a scale-out policy

[Cancel](#) [Create](#)

Step no:7 Now go to the instances page, we can see two instances has been cretaed automatically. Because our desired capcity is 2.

Instances (2) Info

Saved filter sets
Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

☐	Name ✎	Instance ID	Instance state ▼	Instance type ▼	Status check	Al
☐		i-0e0e45b5f3bc04241	Running 🔍 🔍	t3.micro	Initializing 🕒	Vi
☐		i-0fbac5db0ee3e8c20	Running 🔍 🔍	t3.micro	Initializing 🕒	Vi

Try to terminate one instance.

Instances (1/2) Info

Saved filter sets
Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

☐	Name ✎	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm sta	Public IPv
☐		i-0e0e45b5f3bc04241	Running 🔍 🔍	t3.micro	Initializing 🕒	View alar	ec2-15-2C
☒		i-0fbac5db0ee3e8c20	Running 🔍 🔍	t3.micro	Initializing 🕒	View alar	ec2-13-2C

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate (delete) instance

We can see another instance has been created automatically.

Instances (3) Info

Saved filter sets
Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

Last updated less than a minute ago [🔄](#)

☐	Name ✎	Instance ID	Instance state ▼	Instance type ▼	Status check
☐		i-0e0e45b5f3bc04241	Running 🔍 🔍	t3.micro	3/3 checks pass
☐		i-0fbac5db0ee3e8c20	Terminated 🔍 🔍	t3.micro	-
☐		i-082d38386ef3b5306	Running 🔍 🔍	t3.micro	Initializing 🕒

To increase the stress on the instances for checking the scale up **install stress**.

```
[ec2-user@ip-172-31-26-170 ~]$ sudo su
[root@ip-172-31-26-170 ec2-user]# yum install stress
```

Enter the command: **stress -cpu 2 -timeout 300**

It will increase the CPU utilization by 2.

Complete!

```
[root@ip-172-31-26-170 ec2-user]# stress --cpu 2 --timeout 300
stress: info: [25630] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
```

To visualize the utilization enter the following command: **ps -eo pcpu,pid,user,args --sort=-%cpu | head -10**

```
[root@ip-172-31-26-170 ec2-user]# ps -eo pcpu,pid,user,args --sort=-%cpu | head -10
%CPU      PID  USER      COMMAND
 0.2         1  root  /usr/lib/systemd/systemd --switched-root --system --deserialize=32
 0.0         2  root      [kthreadd]
 0.0         3  root      [rcu_gp]
 0.0         4  root      [rcu_par_gp]
 0.0         5  root      [slub_flushwq]
 0.0         6  root      [netns]
 0.0         7  root      [kworker/0:0-cgroup_free]
 0.0         8  root      [kworker/0:0H-events_highpri]
 0.0         9  root      [kworker/u4:0-flush-259:0]
[root@ip-172-31-26-170 ec2-user]#
```

Now go to the instance. As the instance capacity has been met the target the auto scaling will be happened and it will increase the capacity to Maximum capacity as Three.

Instances (4) [Info](#)



Saved filter sets

Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

☐	Name 🔗	Instance ID	Instance state	Instance type	Status check
☐		i-0e0e45b5f3bc04241	🟢 Running 🔍 🔍	t3.micro	🟢 3/3 checks passed
☐		i-0fbac5db0ee3e8c20	⚪ Terminated 🔍 🔍	t3.micro	-
☐		i-082d38386ef3b5306	🟢 Running 🔍 🔍	t3.micro	🟢 3/3 checks passed
☐		i-09adb23719a17e3a8	🟢 Running 🔍 🔍	t3.micro	🕒 Initializing

Note: In order to terminate the instances we must delete the auto scaling policy first.